

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE**IV Semester MCA C1 Test, June - 2026****Subject: CLOUD SECURITY****Time: 75 Minutes****Max. Marks: 30****PART A****Answer the following****2 × 10 = 20**

- 1 a. Define cloud computing. List and explain the five essential characteristics of cloud computing as defined by NIST. CO1 6
- b. Differentiate between IaaS, PaaS, and SaaS with suitable real-world examples for each. CO1 4

OR

- 2 a. Explain the Cloud Computing Architecture in detail. Distinguish between frontend and backend components, and describe each architectural layer. CO1 6
- b. What is the IT foundation for cloud computing? How did traditional IT infrastructure evolve into cloud-based systems? CO1 4

- 3 a. What is Cloud Security Architecture? List its key components and explain how it differs from traditional security architecture. CO2 6
- b. Explain the legal and regulatory issues in cloud computing. Why is data residency and jurisdiction a challenge in a cloud environment? CO2 4

OR

- 4 a. Discuss the key strategies for secure cloud operation. Cover aspects like data encryption, access control, audit logging, and incident response planning. CO2 6
- b. Define security patterns in cloud computing. Give two examples of commonly used security patterns and their purpose. CO2 4

PART B**Answer the following****2 × 5 = 10**

- 5 Illustrate the Cloud Reference Architecture. Identify the roles of Cloud Service Provider, Cloud Consumer, Cloud Broker, Cloud Auditor, and Cloud Carrier. CO1 5

OR

- 6 With a neat diagram, explain the Security Control over the Cloud Model. How does security responsibility vary across IaaS, PaaS, and SaaS? CO1 5

- 7 Compare on-premise security architecture with cloud security architecture. What new risks emerge in the cloud and what architectural patterns are used to mitigate them? CO2 5

OR

- 8 "Legal and regulatory compliance is one of the biggest challenges for organizations moving to the cloud." Discuss covering GDPR, data sovereignty, and compliance frameworks like ISO 27017. CO2 5

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ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE

IV Semester MCA C1 Test, June - 2026

Subject: Block Chain Technology

Time: 75 Minutes

Max. Marks: 30

PART A

Answer the following

2 × 10 = 20

- I 1 Explain the steps involved in RSA key pair generation

CO1 10

OR

- 2 Illustrate with a diagram Point addition in Elliptic Curve Cryptography.

CO1 10

- II 3 What is wallet in bitcoin? Explain in detail about the different types of wallet with example.

CO2 10

OR

- 4 Explain transaction lifecycle in a bitcoin system.

CO2 10

PART B

Answer the following

2 × 5 = 10

- III 5 With a neat diagram explain the generic structure of a block.

CO1 5

OR

- 6 Explain the benefits and limitations of blockchain technology.

CO1 5

- IV 7 What are the types of accounts in Ethereum? Explain briefly.

CO2 5

OR

- 8 Explain the concept of decentralization and its significance in blockchain.

CO2 5
